STAT 293 Final Project

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```r
load("housing.rdata")
dat <- housing_data
```

```r
# ==========================================================
# Missing Data Summary
# ==========================================================
# install.packages("naniar")

miss_var_summary(dat)
```

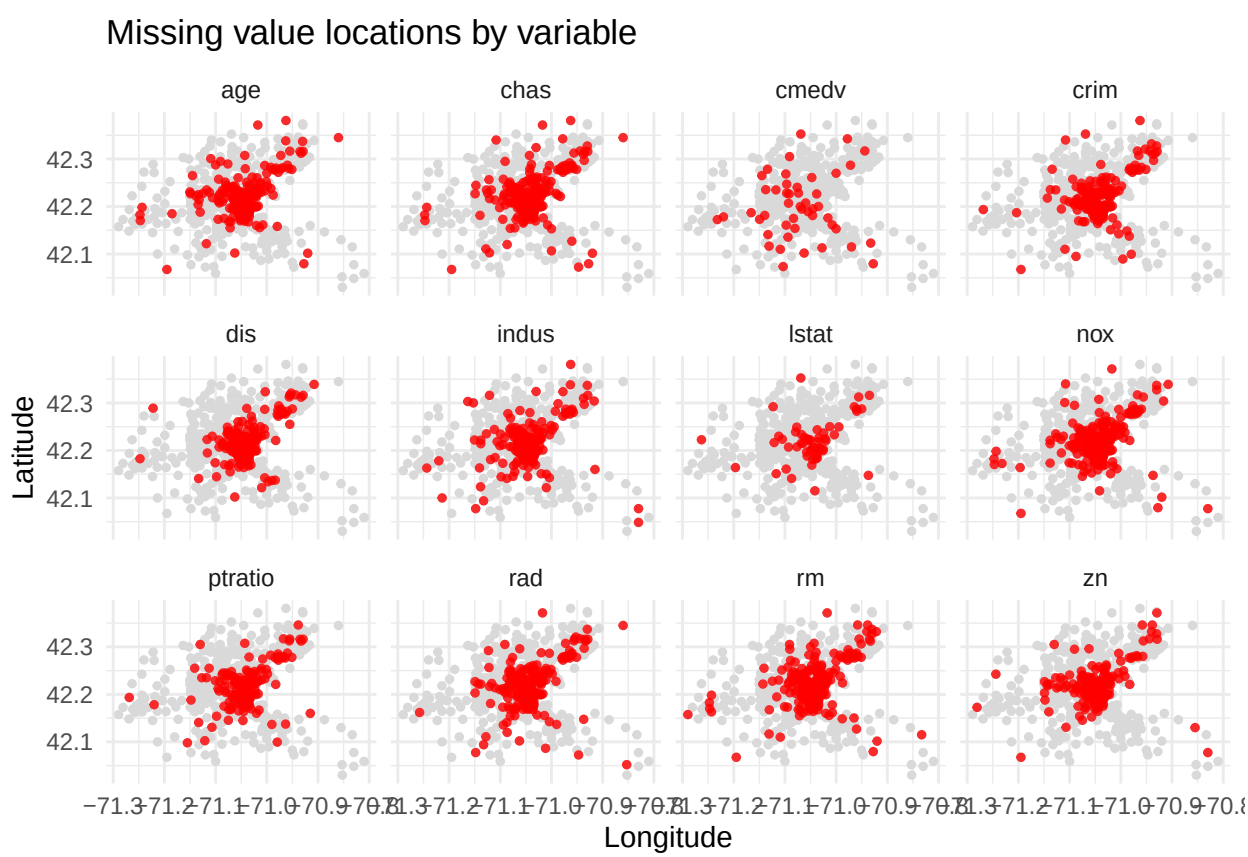
```
## # A tibble: 19 x 3
##    variable n_miss pct_miss
##    <chr>     <int>    <num>
##  1 rm          229    45.3
##  2 rad         219    43.3
##  3 nox         217    42.9
##  4 chas        210    41.5
##  5 age         189    37.4
##  6 indus       180    35.6
##  7 crim        172    34.0
##  8 ptratio     171    33.8
##  9 zn          170    33.6
## 10 dis         165    32.6
## 11 lstat        56    11.1
## 12 cmedv        47     9.29
## 13 town          0     0
## 14 tract         0     0
## 15 lon           0     0
## 16 lat           0     0
## 17 medv          0     0
## 18 tax           0     0
## 19 b             0     0
```

```r
# ==========================================================
# Plot Missing Data by Locations
# ==========================================================
vars_to_check <- c("cmedv", "crim", "zn", "indus", "chas", "nox",
                   "rm", "age", "dis", "rad", "ptratio", "lstat")

missing_long <- dat %>%
  select(lon, lat, all_of(vars_to_check)) %>%
  mutate(across(all_of(vars_to_check), ~ is.na(.))) %>%
  pivot_longer(
```

```r
    cols = all_of(vars_to_check),
    names_to = "variable",
    values_to = "is_missing"
  )

ggplot(missing_long, aes(x = lon, y = lat)) +
  geom_point(color = "grey85", size = 1) +
  geom_point(
    data = subset(missing_long, is_missing),
    color = "red",
    size = 1,
    alpha = 0.8
  ) +
  facet_wrap(~ variable) +
  labs(
    title = "Missing value locations by variable",
    x = "Longitude",
    y = "Latitude"
  ) +
  theme_minimal()
```




```r
# ==========================================================
# 1. Setup imputation
# ==========================================================
init <- mice(dat, maxit = 0)
meth <- init$method
```

```r
pred <- init$predictorMatrix

# variable-specific imputation methods
meth["chas"]  <- "logreg"   # binary
meth["rad"]   <- "polyreg"  # categorical/discrete
meth["cmedv"] <- "pmm"      # continuous target

# remove ID-like variables and medv
pred[, c("town", "tract", "medv")] <- 0

# run MI
imputed_data_set <- mice(
  dat,
  m = 20,
  method = meth,
  predictorMatrix = pred,
  maxit = 10,
  seed = 123
)
```

```
##
##  iter imp variable
##   1   1  cmedv  crim  zn  indus  chas  nox  rm  age  dis  rad  ptratio  lstat
##   1   2  cmedv  crim  zn  indus  chas  nox  rm  age  dis  rad  ptratio  lstat
##   1   3  cmedv  crim  zn  indus  chas  nox  rm  age  dis  rad  ptratio  lstat
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##   2   13  cmedv  crim  zn  indus  chas  nox  rm  age  dis  rad  ptratio  lstat
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## 8 1 cmedv crim zn indus chas nox rm age dis rad ptratio lstat
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```

```r
# ==========================================================
# 2. SAR model
# ==========================================================

# for W matrix
# Extract coordinates
coords <- cbind(dat$lon, dat$lat)

# Define neighbors (using a distance threshold or k-nearest)
ids <- knearneigh(coords, k = 5) # 5 nearest neighbors
nb <- knn2nb(ids)

# Calculate distances for those neighbors
dist_list <- nbdists(nb, coords)

# Apply Inverse Distance function: w = 1/d
inv_dist_list <- lapply(dist_list, function(x) 1/x)

# Convert to a listw object (style="W" performs Row Normalization)
# Row normalization ensures the sum of weights for each row equals 1
listw <- nb2listw(nb, glist = inv_dist_list, style = "W")

fit_sar_model <- function(data) {
  lagsarlm(
    cmedv ~ crim + zn + indus + chas + nox + rm + age +
      dis + rad + tax + ptratio + b + lstat,
    data = data,
    listw = listw
  )
}

# ==========================================================
# 3. Rubin pooling for vector parameters
#    This pools all coefficients jointly, including rho
# ==========================================================
pool_vector_rubin <- function(est_mat, var_mat_list) {
  m <- ncol(est_mat)
  p <- nrow(est_mat)

  qbar <- rowMeans(est_mat)

  # average within-imputation covariance
  ubar <- Reduce(`+`, var_mat_list) / m

  # between-imputation covariance
  b <- matrix(0, p, p)
  for (i in 1:m) {
```

```r
    d <- est_mat[, i] - qbar
    b <- b + tcrossprod(d)
  }
  b <- b / (m - 1)

  # total covariance
  tmat <- ubar + (1 + 1 / m) * b
  se <- sqrt(diag(tmat))

  out <- data.frame(
    term = rownames(est_mat),
    estimate = qbar,
    std.error = se,
    row.names = NULL
  )
  out
}

# ==========================================================
# 4. Primary MAR analysis
# ==========================================================
sar_analyses_mar <- lapply(1:imputed_data_set$m, function(i) {
  single_set <- complete(imputed_data_set, i)
  fit_sar_model(single_set)
})

beta_mat_mar  <- sapply(sar_analyses_mar, coef)
vcov_list_mar <- lapply(sar_analyses_mar, vcov)

pooled_beta_mar <- pool_vector_rubin(beta_mat_mar, vcov_list_mar)

cat("\n--- Primary MAR pooled results ---\n")
```

```
##
## --- Primary MAR pooled results ---
```

```r
print(pooled_beta_mar)
```

```
##           term      estimate   std.error
## 1          rho   0.294217745 0.047197584
## 2  (Intercept) -11.417022010 5.758545817
## 3         crim  -0.437972170 0.094296102
## 4           zn   0.044269928 0.013573948
## 5        indus   0.164506404 0.074798349
## 6        chas1   0.222005258 0.874234048
## 7          nox  -8.477965550 5.898954511
## 8           rm   6.909474213 0.682135980
## 9          age  -0.015578681 0.017158983
## 10         dis  -0.771683929 0.219858846
## 11        rad2   0.181466291 1.270212712
## 12        rad3   3.915684742 1.060046074
## 13        rad4   2.602490955 0.909631353
## 14        rad5   2.600110462 1.003138128
## 15        rad6   0.888707872 1.345356845
```

```
## 16        rad7   3.161562878 1.252356405
## 17        rad8   3.408737514 1.210011850
## 18       rad24   6.069892453 1.668263669
## 19         tax  -0.008203083 0.004642973
## 20     ptratio  -0.501885526 0.141725369
## 21           b   0.006836138 0.003003582
## 22       lstat  -0.178023911 0.068293876
```

```r
cat("\n--- Pooled rho under MAR ---\n")
```

```
##
## --- Pooled rho under MAR ---
```

```r
print(subset(pooled_beta_mar, term == "rho"))
```

```
##   term  estimate  std.error
## 1  rho 0.2942177 0.04719758
```

```r
# ==========================================================
# 5. Sensitivity analysis: delta-adjusted pattern-mixture
#    Shift only originally missing cmedv values
# ==========================================================
miss_cmedv <- is.na(dat$cmedv)

apply_delta_cmedv <- function(data_completed, miss_index, delta) {
  data_new <- data_completed
  data_new$cmedv[miss_index] <- data_new$cmedv[miss_index] + delta
  data_new
}

# choose delta values using SD of observed cmedv
sd_cmedv <- sd(dat$cmedv, na.rm = TRUE)
deltas <- c(-0.3, -0.1, 0, 0.1, 0.3) * sd_cmedv

sens_results_beta <- vector("list", length(deltas))
names(sens_results_beta) <- as.character(deltas)

for (dd in deltas) {
  sar_analyses_delta <- lapply(1:imputed_data_set$m, function(i) {
    single_set <- complete(imputed_data_set, i)
    single_set_delta <- apply_delta_cmedv(single_set, miss_cmedv, dd)
    fit_sar_model(single_set_delta)
  })

  beta_mat_delta  <- sapply(sar_analyses_delta, coef)
  vcov_list_delta <- lapply(sar_analyses_delta, vcov)

  pooled_beta_delta <- pool_vector_rubin(beta_mat_delta, vcov_list_delta)
  pooled_beta_delta$delta <- dd

  sens_results_beta[[as.character(dd)]] <- pooled_beta_delta
}

sens_beta_tab <- rbindlist(sens_results_beta, fill = TRUE)

cat("\n--- Sensitivity analysis pooled results ---\n")
```

```
##
## --- Sensitivity analysis pooled results ---
```

```r
print(sens_beta_tab)
```

```
##             term      estimate   std.error     delta
##           <char>         <num>       <num>     <num>
##   1:         rho   0.285171232 0.049362935 -2.760533
##   2: (Intercept) -13.105929474 6.031390811 -2.760533
##   3:        crim  -0.440499551 0.095134608 -2.760533
##   4:          zn   0.045158574 0.014122355 -2.760533
##   5:       indus   0.172033889 0.076674921 -2.760533
##  ---
## 106:       rad24   6.340400927 1.698323264  2.760533
## 107:         tax  -0.007834891 0.004669927  2.760533
## 108:     ptratio  -0.528314125 0.143029739  2.760533
## 109:           b   0.006657239 0.002969889  2.760533
## 110:       lstat  -0.191262575 0.066791917  2.760533
```

```r
cat("\n--- Sensitivity analysis: rho only ---\n")
```

```
##
## --- Sensitivity analysis: rho only ---
```

```r
print(subset(sens_beta_tab, term == "rho"))
```

```
##      term  estimate  std.error      delta
##    <char>     <num>      <num>      <num>
## 1:    rho 0.2851712 0.04936293 -2.7605325
## 2:    rho 0.2923188 0.04785953 -0.9201775
## 3:    rho 0.2942177 0.04719758  0.0000000
## 4:    rho 0.2949455 0.04661821  0.9201775
## 5:    rho 0.2928524 0.04576430  2.7605325
```